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ASSESSING A DECOMMISSIONED LANDFILL USING ELECTRICAL RESISTIVITY TOMOGRAPHY (ERT)

CASE STUDY: OTI LANDFILL, DOMPOASE

BY
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Declaration

I hereby declare that this submission is my own work towards the award of the Bachelor of Science degree in Civil Engineering and that, to the best of my knowledge, it contains no material previously published by another person, nor material which has been accepted for the award of any other degree of the University, except where due acknowledgment has been made in the text.
I further declare that no part of this work has been submitted to any other institution for the award of a degree or diploma.
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Abstract

Rapid urbanisation and population growth in Ghana have heightened the environmental risks associated with municipal solid waste disposal, especially in landfills that lack engineered containment systems. The Oti Landfill in Dompoase, which was originally located away from residential areas, now sits adjacent to urban settlements. This proximity raises concerns about leachate infiltration and gas emissions from its decommissioned cells.

This study utilises Electrical Resistivity Tomography (ERT) to non-invasively evaluate the integrity of the capping layer and liner system within the decommissioned section of the landfill. Field surveys conducted in May 2025 aimed to determine the thickness and lateral uniformity of the cap, identify subsurface anomalies indicative of leachate and gas hotspots, and estimate the depth of the liner.

The resistivity profiles revealed zones of low resistivity (<10Ω·m), suggesting potential leachate accumulation and compromised containment. These findings highlight the value of geophysical methods in landfill diagnostics and provide critical insights for post-closure monitoring, environmental risk mitigation, and future site redevelopment planning.

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1 Introduction

1.1 Background

Human activities, from manufacturing and processing to consumption, inevitably produce waste. Managing municipal solid waste (MSW) poses a major environmental and public health challenge for large cities in developing countries (Renou et al., 2008, cited in Boateng et al., 2019). Among various waste management strategies, landfilling is often the simplest and most cost-effective method for solid waste. Developing countries mainly depend on landfills for waste disposal because they offer significant advantages, such as low technological demands and economic efficiency (Gonzalez-Valencia et al., 2016). However, many landfills in these countries operate below acceptable standards (Oyeku and Eludoyin, 2010), and waste is frequently not sorted, which can lead to the accidental release of leachates and toxic gases into the environment (Alimba et al., 2012; Schrapp and Al-Mutairi, 2010). It is common to find unhygienic landfills close to residential areas in such nations (Alimba et al., 2006). Leachates form when waste contacts water that seeps into landfill sites and can leak through liners, contaminating soil and aquatic environments due to manufacturing or construction defects, as well as vapour diffusion through liners or capping layers (Grugnaletti et al., 2016; Pantini et al., 2014). As a result, areas near landfills face increased water pollution risks, affecting both groundwater and surface water, due to potential leachate seepage from these sites. This uncontrolled release can cause significant environmental and health problems in many developing countries (Adamcová et al., 2017; Oyeku and Eludoyin, 2010; Samadder et al., 2017). Given the high waste generation rates and inadequate infrastructure, managing solid waste has become a pressing issue for densely populated developing nations. The municipal waste problem is especially critical in countries like Ghana, where waste disposal mainly involves dump yards or landfills lacking proper leachate collection and treatment systems. In Ghana, waste volume has surged alongside rapid population growth and urban, commercial, and industrial expansion. Moreover, waste management systems for industrial and urban waste are limited, with many landfills, often used as disposal sites, located in urban rivers, agricultural lands, and public spaces without treatment measures in place. In Ghana, the composition of solid waste varies greatly, as all types, including food waste, batteries, construction materials, plastics, and others, are discarded together without proper sorting. The Oti Landfill, located in Dompoase, has been used for waste disposal since 2004. (Fig 1) At first, the landfill was situated away from residential areas; however, due to urbanisation and population growth, many homes now lie close to the site. Portions of the landfill have been capped and decommissioned over recent years. Some of the cells in the decommissioned section were lined with geomembrane. Over time, the waste decomposes, generating leachate and gases within the landfill. If landfill liners fail, due to construction defects, ageing, or vapour diffusion, leachate can infiltrate surrounding soils and contaminate nearby water bodies. In parallel, the decomposition of waste generates gases that may pose environmental and public health risks if not properly managed. Therefore, a comprehensive assessment of landfill components, such as liners and capping layers, is essential to determine whether they are functioning as intended. Such evaluations are critical for safeguarding groundwater, surface water, and ecological integrity in urban environments.

1.2 Objective

This study uses electrical resistivity methods to evaluate the condition of the capping layer and liner in the decommissioned sections of the Oti Landfill. The focus will be on:

- Determining the thickness of the capping layer
- Assessing the lateral uniformity of the capping layer
- Characterising the landfill waste to identify hotspot zones where leachate and gases are likely to be generated.
- Determining the depth of the liner

1.3 Justification

The closeness of residential areas to the Oti Landfill and the potential risk of leachate infiltration and gas emissions from compromised liners. Employing non-invasive, cost-effective methods such as Electrical Resistivity can be beneficial. Additionally, the findings obtained from this study will be important for decisions regarding future site redevelopment.

1.4 Scope

This study uses the Electrical Resistivity method to assess the decommissioned section of the Oti Landfill in Dompoase. It aims to identify subsurface anomalies, indicative of leachate and hotspot zones for gas production. The scope is limited to geophysical characterisation and does not include chemical analysis of leachate or air quality monitoring. Field surveys were conducted in May 2025, and data interpretation was based on resistivity profiles and site observations.

Figure 1: The Oti Landfill, the study area

2 Literature Review

2.1 Landfill

A landfill is a specially designed site where solid waste is stored until it is deemed safe. *Safety* in this context is generally defined as the point at which the waste no longer threatens human health or the environment, usually after it has decomposed or stabilised over time. Landfills have a planned lifespan, including active and inactive (closure) periods. According to Chaitanaya *et al.* (2017), in their paper, *Impact on Environment by Landfill and A Case Study on Vishakhapatnam Landfill*, the active period typically lasts between 10 and 25 years, depending on land availability. Once the active period concludes, the landfill is monitored and maintained for an additional 25 years during the closure period.
In a conventional sanitary landfill, waste is kept somewhat dry by using a base liner and cover. This design prevents the infiltration of water from the landfill. The liquid resulting from the biodegradation of the waste, known as leachate, is collected at the bottom of the landfill and disposed of securely. (Fig 2) The containment and restriction of liquid infiltration can slow down the degradation of MSW, leading to less complete and less predictable degradation that can persist for many years. (Sharma and Reddy, 2004, cited in Hettiaratchi *et al*., 2016)

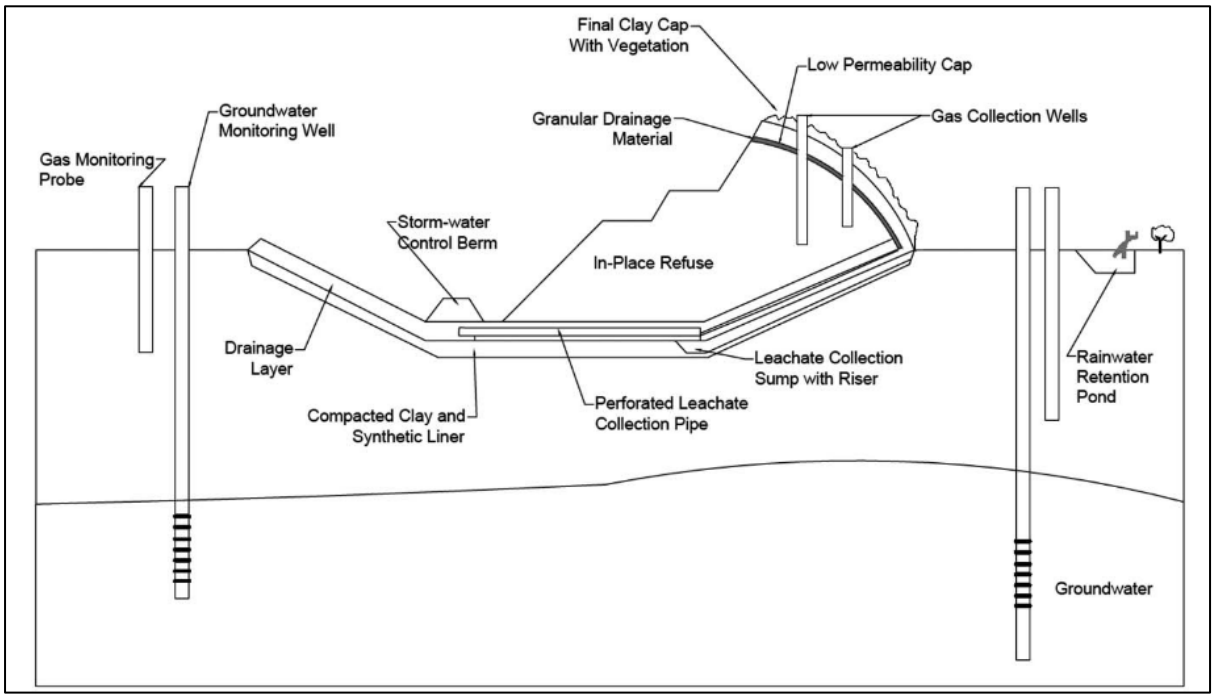

Figure 2: A cross-section of an engineered landfill

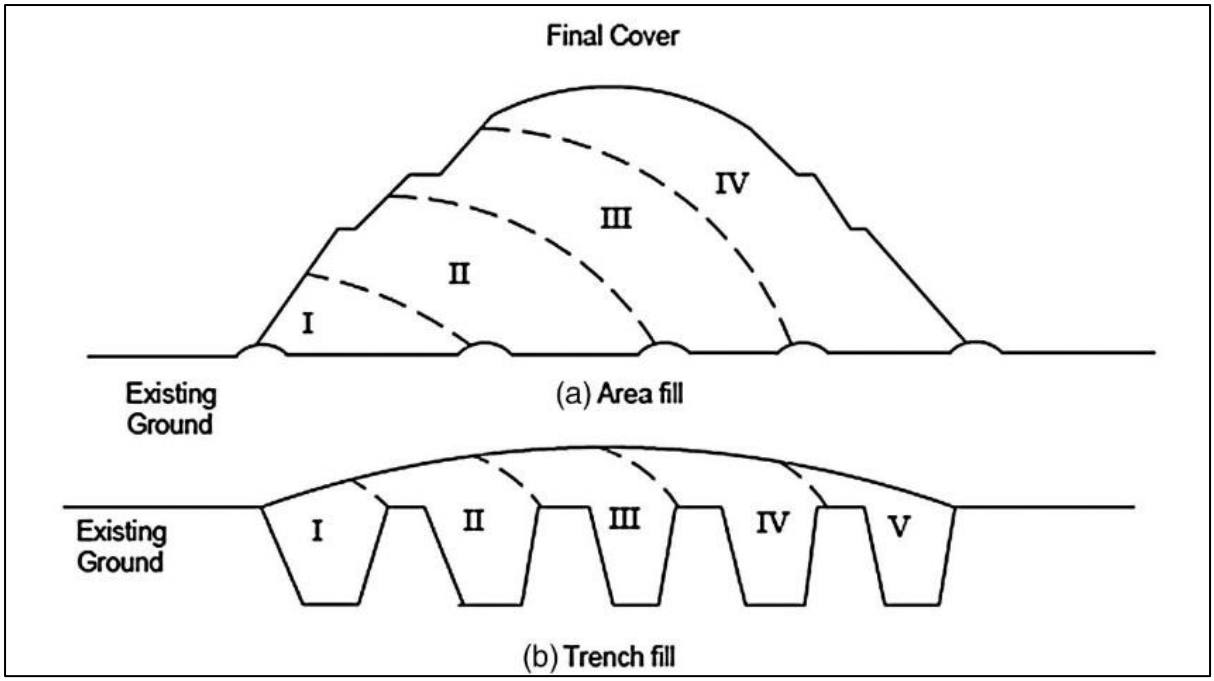

Figure 3: A typical Landfill configuration

2.1.1 Types of Landfills

- Sanitary Landfills: They are specially designed facilities for the safe disposal of MSW. They include features such as liners, leachate collection systems, and daily soil cover to minimise their environmental impact. These landfills effectively isolate waste from groundwater and air, which helps reduce the risks of contamination and disease. (Hettiaratchi *et al*., 2016)
- Hazardous Waste Landfills: These landfills are designed to contain toxic industrial waste, including chemicals, heavy metals, and other hazardous materials. They use double-liner systems, leachate detection layers, and strict access controls to prevent environmental release. (Qian, X. *et al*., 2002)
- Bioreactor landfills: They accelerate waste decomposition by actively managing moisture and air content. Leachate is recirculated to stimulate microbial activity, enhancing gas production and reducing landfill lifespan. (Reinhart *et al*., 1998)
- Industrial Waste Landfill: They are used for non-hazardous industrial by-products like fly ash, foundry sand, and sludge. Design varies based on waste characteristics, but liners and monitoring systems are standard. (Tchobanoglous *et al*., 1993)

2.1.2 Components of a Landfill

- Cover (Cap): The cover is compacted soil or some other material that is placed over the waste in a cell after it is full.
- Cells: These are the spaces where the wastes are disposed of.
- Bottom liner system (Barrier): This serves as a separation between the waste and the outside soil.
- Leachate collection system: That which collects leachates to the leachate collection pond.

2.1.3 Landfill Liner System

A landfill liner is a low-permeability barrier system placed at the base of a landfill to prevent leachate, formed from decomposing waste, from migrating into the surrounding soil and groundwater. It is one of the most critical components in engineered landfills, particularly for sanitary and hazardous waste landfills.

There are two major types of landfill liner systems: the Single Liner and the Double Liner. According to Solomon, A., and Poulose, E. (2018), Single liners are typically used in landfills that mostly contain construction rubble. These landfills are not designed to hold harmful liquid wastes, such as paint or tar, which can easily seep through a single liner system. On the other hand, double liners are commonly found in municipal solid waste landfills as well as hazardous waste landfills. The first layer of a double liner collects the leachate, while the second layer serves as a leak-detection system, ensuring that no contaminants escape into the ground and cause pollution.

Landfill liners are primarily made from various materials, with clay and geosynthetics being the most common. According to Das and Sohban (2013), clay was the dominant liner material used in landfills until around 1982. A well-compacted clay liners typically have a hydraulic conductivity of about 10^{-7}cm/sec or lower. Although geosynthetics have been utilised in civil engineering since the late 1970s, they were only later incorporated into landfill design.

Geosynthetics, particularly geomembranes, are commonly used materials for constructing landfill liners. Geomembranes are flexible barriers that are impermeable to liquids or vapours, primarily made from continuous sheets of polymer. The polymeric materials used for geomembranes can be classified as either thermoplastic or thermoset. Thermoplastic polymers include PVC (polyvinyl chloride), polyethylene, chlorinated polyethylene, and polyamide. In contrast, thermoset polymers consist of ethylene vinyl acetate, polychloroprene, and isoprene-isobutylene. While geomembranes are generally considered impermeable, they are not completely so. Water vapour transmission tests indicate that the hydraulic conductivity of geomembranes ranges from 10^{-10} to 10^{-13} cm/sec, which means they are only essentially impermeable. (Das and Sohban, 2013)

2.1.4 Leachates

Leachate is a contaminated liquid that forms when water seeps through waste materials in a landfill, dissolving and carrying various pollutants, including organic and inorganic substances such as heavy metals, salts, and pathogens.

Leachate primarily forms when waste contacts with water, but its volume and composition are influenced by factors such as the landfill's age, engineering design, the types of waste present, and prevailing climatic conditions (Srivastava *et al*., 2025). Leachate comprises various constituents, including dissolved organic matter, heavy metals, ammonia, and organic salts. The dominance of these components can vary depending on the waste composition in the landfill. Additionally, the type of waste influences the level of biological activity within the landfill. For instance, rubbish, food scraps, garden waste, and animal waste contribute to the organic portion of leachate, whereas construction and demolition debris add to the inorganic components. (Al Raisi, 2022)

Landfill leachates are chemically complex fluids, and their electrical properties, especially electrical conductivity (EC) and resistivity, provide essential insights into the dynamics of subsurface contamination. The EC of leachate typically ranges from 2,000 to over 15,000 µS/cm, which indicates high concentrations of dissolved ions such as chlorides, sulfates, ammonia, and heavy metals (Srivastava *et al.*, 2025; Alao *et al*., 2024). These elevated levels of ions lead to low resistivity zones of less than 10 Ω·m.

2.1.5 Landfill Closure

Landfill capping systems are integral to the closure phase of waste disposal sites, serving both environmental and structural functions. Their primary objectives include minimising water infiltration, promoting surface drainage, controlling landfill gas migration, and providing a protective barrier between the waste and surrounding ecosystems. The effectiveness of these systems directly influences the environmental, health, and economic outcomes associated with the site (Strata Geosystems, 2023; MIT OCW, 2004). Common capping materials include compacted soil, geosynthetics, and vegetation, selected based on site-specific conditions and

long-term performance goals (Ramalho *et al*., 2013; IntechOpen, 2023). A natural compacted soil layer should have a minimum thickness of 0.6m and a hydraulic conductivty not exceeding $10^{-9}\,m/s$. Alternatively, geosynthetic materials may be used if they offer equivalent protection.

One of the major engineering challenges in landfill capping is settlement, which results from the decomposition of biodegradable waste. MSW landfills typically experience settlement values ranging from 10% to 25% of the total waste height. To accommodate differential settlement and maintain cap integrity, systems may be designed with increased thickness or slope. When geomembranes are used, they must withstand tensile strains induced by uneven subsidence. Materials such as linear low-density polyethene (LLDPE) are particularly suitable due to their flexibility and durability. Despite careful design, post-closure maintenance may still be required, including regrading of the final cover to restore surface stability and drainage function (EPA, 1993). Properly engineered caps also play a vital role in minimising leachate production and protecting groundwater resources by reducing percolation through the waste mass. (Chen *et al*., 2012, cited in Manrique *et al*., 2019)

Additionally, they serve as primary controls for landfill gas emissions, particularly methane, which poses both explosive and climate risks. Advanced designs may incorporate gas venting layers that facilitate safe dissipation and enable methane capture for flaring or energy recovery, transforming a liability into a renewable energy asset. (Wemegah, 2015; MDPI, 2024)

Beyond containment, landfill caps support post-closure land-use by stabilising the surface and enabling site reclamation. Capped landfills can be repurposed for recreational parks, commercial developments, or green spaces, enhancing community value and aesthetics while promoting economic redevelopment (NumberAnalytics, 2023). In this way, engineered capping systems not only safeguard environmental integrity but also unlock the potential for sustainable land transformation.

2.2 Application of Geophysical Methods in Engineering

Geophysical methods are techniques based on physics used in various fields to study the Earth's subsurface. These methods are essential in areas such as civil engineering, environmental engineering, archaeology, planetary exploration, and more.

In civil and environmental engineering, geophysical techniques are applied across various areas, such as construction site assessments, inspections of dams and dikes, pavement inspections and environmental investigations. The main advantage of these techniques is their ability to efficiently and non-invasively detect subsurface features, which eliminates the need for intrusive procedures like coring and drilling or borehole excavation. These methods provide a comprehensive and often detailed view of subsurface conditions while allowing for rapid and cost-effective data collection. However, their effectiveness is constrained by factors such as resolution limits, environmental noise, and the non-uniqueness of inversion models, which can affect accuracy in interpretation. Selecting an appropriate geophysical method requires careful consideration of several factors. First, the specific objectives of the study (e.g., groundwater detection, rock characterisation, soil profiling, leachates, or cavity identification). Second, the physical properties being measured. Third, the methodological configuration must align with the investigation's spatial resolution and depth requirements. Environmental conditions, such as site noise, moisture variability, and cultural interference, can further influence data quality. Geophysical surveys should be calibrated with ground truth data and informed by existing literature or previously published studies relevant to the site to enhance reliability. (Hermann J. Mauritsch, 2000; Rhett Herman, 2001; Soupios *et al*., 2007; Mondelli *et. al*, 2013; Osinowo *et. al*, 2019), cited in Iman Farshchi *et al*. (2025). Building on these foundational principles, it is essential to understand how specific geophysical techniques align with engineering targets in real-world applications. The selection of a method depends not only on the physical properties being measured but also on the nature of the subsurface feature, the required

resolution, and the environmental context. Table 1 summarises key engineering targets and the geophysical methods commonly employed for their direct detection and structural interpretation, including landfill leachate contamination, a central focus of this study.

Table 1: Geophysical Methods in Engineering Applications

Engineering Target	**Geophysical Techniques**
Overburden (soil, fill, weathered rock)	Resistivity, Seismic
Location of pipes or conduits	Electromagnetics, GPR, Resistivity
Nondestructive material evaluation	Electromagnetics, Resistivity
Sediments (engineering site assessment)	Resistivity, Seismic
Geothermal studies (thermal gradients)	Resistivity, Audio – Magnetotellurics
Asbestos or hazardous material zones	Electromagnetics, Radiometrics
Ore deposits or buried objects	Electromagnetics, GPR, Resistivity
Landfill Leachate Contamination	*Resistivity, IP, Electromagnetics*

According to Hazreek *et al*. (2011), techniques such as Ground Penetrating Radar (GPR) and seismic methods have shown potential for providing information related to pavement layer thickness and depth. These methods are particularly useful for long continuous assessments where traditional coring tests may be impractical or damaging to the pavement structure. Among these geophysical techniques, seismic methods have proven to be especially effective in geotechnical and engineering geology contexts, offering reliable estimates of bedrock depth, material type, and layer thickness. Their use not only reduces the need for destructive sampling but also improves spatial coverage and interpretive confidence when combined with ground truth data.

Geophysical methods have played a pivotal role in the evolution of petroleum exploration, transitioning from early surface-based observations to sophisticated subsurface imaging techniques. Historically, wildcatters located oil by drilling near natural seeps and prominent geological structures such as anticlines in exposed rock formations. However, as these obvious prospects were quickly exhausted, geologists adopted seismic surveys to detect more subtle subsurface traps. Seismic technology, initially used in the early 1900s for measuring water depths and detecting icebergs, was first applied to oil exploration in 1924 during the discovery of a Texan oil field (Milligan, 2004, cited in Amadike *et al*., 2024). The introduction of three-dimensional (3D) seismic methods marked a significant advancement, enabling detailed reservoir visualisation and substantially reducing exploration risks. As Aminzadeh and Dasgupta emphasised in their book, *Geophysics for Petroleum Engineers*, geophysical techniques contribute not only to reservoir characterisation and monitoring but also to strategic reservoir management. These methods enhance production planning and mitigate risks such as dry wells, blowouts, and inefficient recovery. By providing scientifically grounded interpretations of subsurface conditions, geophysics serves as both a value-adding and risk-reduction tool in petroleum ventures, guiding critical business decisions and optimising resource development.

One of the principal risks associated with small-scale mining is the lack of sufficient geological information to support efficient mine planning, which often leads to operational failures and financial losses. This limitation is largely attributed to financial constraints that prevent small-scale operators from conducting detailed geological surveys or investing in conventional exploration techniques (Hruschka and Echavarria; Hentschel *et al*.; Abichequer *et al*.; Godoy, cited in Tichauer *et al*., 2020). In response to these challenges, geophysical methods have emerged as cost-effective alternatives for reducing geological uncertainty and enhancing decision-making. As summarised by Tichauer *et al*. (2020), geophysical techniques, particularly electrical, electromagnetic, gravimetric, magnetic, and seismic methods, are widely applied in mineral exploration to identify and delineate subsurface deposits. Shallow

geophysics is especially useful for investigating near-surface geological structures with limited spatial extent. Within the same study, Martins *et al*. are cited for demonstrating that resistivity and induced polarisation (IP) surveys can significantly improve geological modelling in small-scale mining contexts, offering rapid and low-cost data acquisition while reducing the number of unsuccessful drill holes by 30–50%. Consequently, the integration of geophysical methods throughout all phases of a small-scale mining project, from exploration to planning, can substantially improve deposit delineation and support informed decision-making, thereby mitigating risk and optimising resource utilization. (Tichauer *et al*., 2020)

2.2.1 Application of Geophysical Methods in Landfills

The characterisation stage is essential for evaluating a landfill's potential for waste-to-material or waste-to-energy recovery. Conventional investigation techniques, such as borehole drilling and trial pit excavation, are invasive, time-consuming, and costly. They may also introduce new pathways for contaminant migration and yield results that are spatially limited and poorly representative of the landfill's true heterogeneity (De Carlo *et al.*, 2013; Jones *et al.*, 2013; Dumont *et al.*, 2017, cited in Manrique *et al.*, 2019). These limitations have prompted the adoption of non-invasive geophysical methods, which offer a cost-effective and environmentally sustainable alternative for subsurface characterisation. Advances in geosciences have enabled these techniques to target a wide range of landfill features, from delineating waste boundaries to assessing hydrodynamic behaviour, while minimising site disturbance (Nguyen *et al.*, 2018, cited in Manrique *et al.*, 2019).

Electrical methods are particularly effective in landfill investigations due to the saline, organic-rich, and electrically conductive nature of leachate, which enhances subsurface detectability. As summarised by Manrique *et al.* (2019), techniques such as electrical resistivity, magnetics, magnetic susceptibility, and time spectral induced polarisation provide rapid, minimally invasive, and high-resolution data at the field scale, making them valuable tools for assessing landfill impacts. Environmental assessments using geophysical methods have been extensively discussed in the literature (Greenhouse and Harris, 1983; Sharma, 1997; Meju, 2000; Monteiro Santos *et al.*, 2002; Dawson *et al.*, 2002, cited in Ramalho *et al.*, 2013). These studies emphasise the complex physicochemical variability of landfills, which stems from the heterogeneous nature of waste materials and deposition conditions (Meju, 2000).

To address this complexity, technologies such as 2D/3D geoelectrical tomography, refraction seismic surveys, electromagnetic profiling, ambient noise measurements, and ground penetrating radar, often combined with on-site analyses, have been employed to estimate subsurface properties. These methods enable the identification of waste layering and stratigraphy, thereby improving the geometric definition of the site under investigation (Soupios *et al.*, 2006, in press; Saltas *et al.*, 2005, cited in Ramalho *et al.*, 2013).

2.2.2 Electrical Resistivity

Electrical resistivity surveys are geophysical techniques used to map subsurface structures by measuring the electrical properties close to the ground surface. This method involves injecting an electric current into the ground through two current electrodes and measuring the resulting voltage difference between two potential electrodes. In this context, resistivity refers to the intrinsic property of a material that quantifies its opposition to the flow of electric current, expressed in ohm-m (Ω·m). The measurements obtained are used to calculate apparent resistivity using the formula:

$$\rho = k \times (\Delta V / I)$$

where ρ represents the apparent resistivity, ΔV is the measured potential difference, I is the injected current, and k is the geometric factor, which depends on the array configuration. The term *apparent resistivity* is used because the resistivity values measured are averages over the total current path length, but they are plotted at one depth point for each pair of potential electrodes (Applied Geophysics, 2018). By varying the positions of the electrodes, a dataset of

apparent resistivity values is generated, which is then inverted to estimate the true subsurface resistivity distribution. Ground resistivity is influenced by various geological factors, including mineral content, fluid presence, porosity, and ionic concentration. (British Geological Survey, n.d.)

Resistivity surveys are typically classified into profiling and sounding methods. Profiling surveys, such as those using the Wenner array, maintain fixed electrode spacing and move the array laterally to detect horizontal variations. In contrast, Vertical Electrical Sounding (VES), commonly implemented with the Schlumberger array, keeps the array centre fixed while increasing electrode spacing to probe deeper layers. VES is particularly effective in groundwater exploration where a one-dimensional (1D) model is acceptable. However, its inability to account for lateral resistivity variations can lead to misinterpretation of layer thickness and composition. For more accurate results, two-dimensional (2D) surveys and inversion models are preferred, especially in complex geological settings. (Meju, 2000)

A more advanced form of resistivity surveying, Electrical Resistivity Tomography (ERT), enhances spatial resolution by deploying multiple electrodes across a grid. This technique produces high-resolution images of subsurface resistivity variations, enabling detailed characterisation of geological features such as fracture zones, contaminant plumes, and landfill boundaries. In landfill investigations, both ERT and VES are highly effective due to their sensitivity to moisture content, ionic concentration, and waste decomposition processes. The presence of salts and organic acids in leachate significantly lowers the resistivity of surrounding soils and aquifers, making these methods valuable for delineating waste boundaries, estimating fill thickness, and detecting leachate-impacted zones. (Meju, 2000)

Despite its widespread use, resistivity surveying faces challenges such as inversion instability and data noise, which can produce non-physical outputs like negative resistivity values. These anomalies typically indicate errors in measurement or model fitting and must be carefully filtered or constrained during data processing to ensure reliable interpretation. (British Geological Survey, n.d.)

2.2.2.1 Array Configuration

Electrode array configuration is crucial for the effectiveness of Electrical Resistivity Tomography (ERT). It affects data resolution, depth of investigation (DoI), and sensitivity to subsurface heterogeneities. In landfill studies, selecting the appropriate array is essential for detecting leachate plumes, delineating waste boundaries, and assessing both vertical and lateral variations in resistivity. There are various types of electrode arrays used in landfill studies, but the most employed are the Wenner, Schlumberger, and Dipole-Dipole arrays. (Fig 4)

The Wenner array uses equal spacing between all four electrodes, making it particularly effective in layered environments. The Schlumberger array, on the other hand, has wider spacing between the current electrodes and closer spacing between the potential electrodes, which makes it suitable for deeper subsurface profiling. The Dipole-Dipole array arranges the current and potential electrodes in pairs with equal spacing, providing high sensitivity to lateral variations in resistivity. Additionally, there are other types of arrays, including the Pole – Pole, Pole – Dipole, and Gradient arrays.

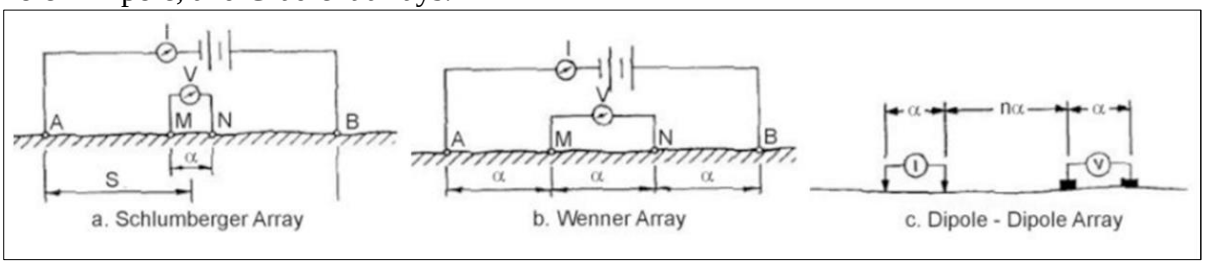

Figure 4: Array configuration

2.2.2.2 Electrode Spacing

The spacing between electrodes in ERT surveys significantly influences the results. Generally, wider electrode spacing allows for a deeper subsurface investigation. However, the choice of electrode spacing can greatly impact the overall ERT model. If electrode spacing is not selected properly, it may lead to an incomplete electrical resistivity model, resulting in inaccurate or misleading interpretations of the subsurface. (Loke, 2021; Abdul-Nafui A.K., 2013, cited in Wangyao, 2023)

For exploring shallow layers, a smaller electrode spacing provides a denser array of apparent resistivity measurements of the subsurface. As electrode spacing increases, the accuracy of the inverted ERT images diminishes concerning both resistivity distribution and the delineation of interfaces. This degradation becomes particularly pronounced when the electrode spacing exceeds the thickness of the top subsurface layer. (Gourdol *et al.*, 2021)

2.2.2.3 Depth of Investigation

Both experimental and field-based studies confirm that the depth of investigation (DoI) in ERT is influenced not only by electrode spacing but also by the type of array used. Asare *et al.* (2024), through laboratory measurements with Schlumberger and Wenner arrays, found that the actual DoI is approximately 0.25 times the current electrode spacing (AB). This finding is inconsistent with the conventional assumption that the DoI is equal to AB/2 and highlights the need for site-specific calibration, particularly in layered or heterogeneous landfill environments. The IRIS Instruments Technical Bulletin also provides practical field guidelines, suggesting that the maximum DoI is typically around 20% of the total cable length. For instance, with a 108 m cable and 4 m electrode spacing, the expected DoI would be about 20 m.

2.2.2.4 Data Processing and Software

Electrical Resistivity Tomography data processing involves several key steps: the acquisition of voltage and current measurements, conversion of these measurements to apparent resistivity values, filtering out noise and outliers, and conducting inversion modelling to reconstruct subsurface resistivity distributions. Inversion algorithms, which are typically based on least-squares or smoothness-constrained methods, generate 2D or 3D resistivity models that reveal subsurface heterogeneity, leachate plumes, and landfill geometry. (Gourdol *et al.*, 2021; Isobe and Ishimori, 2025)

Several software platforms support these workflows. Res2DInv (Geotomo Software) is widely used for the 2D inversion of direct current (DC) resistivity and induced polarisation (IP) data. It accommodates all conventional and custom electrode configurations, including underwater and cross-borehole surveys. The software offers flexible control over inversion parameters, topographic corrections, and the integration of time-lapse data. (Aarhus GeoSoftware, 2025) EarthImager 2D (AGI) provides a user-friendly interface for resistivity and IP inversion. It includes automated bad-data removal, time-lapse monitoring, and outputs that meet high-definition engineering standards. The software supports joint inversion of multiple array types and offers tools for visualising convergence curves and misfit histograms. (AGI, 2021)

2.2.2.5 Causes and Implications of Negative Resistivity

Resistivity values obtained from ERT should ideally be positive, as they indicate the material's resistance to electrical current. However, negative resistivity values can sometimes appear in raw datasets or inversion outputs. These anomalies are non-physical and require careful interpretation.

One specific cause of negative resistivity is the presence of highly conductive materials near the receiving electrodes. For instance, when a conductive zone, such as a leachate plume or a saturated waste pocket, is located close to an electrode, it can act as a secondary source of current. This can distort the expected voltage gradient, potentially reversing the polarity of the measured potential difference. As a result, the inversion algorithm may calculate apparent

negative resistivity values (Guideline Geo, n.d.). Such effects are particularly pronounced in landfill environments, where ionic-rich leachate significantly lowers resistivity and may concentrate along the paths of the electrodes. Other contributing factors include the use of array configurations with large geometric factors, such as dipole-dipole arrays, which amplify small voltage errors and increase sensitivity to noise (RES2DINV Manual, 2016). Electrode contact resistance, particularly in dry or compacted soils, can further distort current flow and voltage readings, introducing instability into the dataset. Additionally, boundary effects within inversion models, especially at poorly constrained edges, may produce unrealistic resistivity estimates due to insufficient data coverage. In urban or industrial settings, electromagnetic interference and data outliers can also degrade signal quality, compounding the risk of inversion artefacts. (SEP Geophysical, n.d.)

In landfill investigations, where resistivity contrasts are sharp due to moisture variability and waste heterogeneity, inversion artefacts can significantly affect interpretation. Misinterpreting negative resistivity zones may lead to incorrect delineation of leachate plumes or waste boundaries. A clear understanding of the physical and computational origins of these anomalies enhances the interpretive reliability of resistivity models, particularly in complex landfill environments.

3 Methodology

3.1 Study Area

Kumasi Metropolis has about 4 million people, owing to travellers from neighbouring districts who come to conduct business activities. The waste produced in Kumasi originates from households, industrial, institutional and commercial sources. The city supports a wide range of products, which include plastics, textiles, telecommunication components, electronics, construction and biomedical products.

The Oti Landfill is an engineered sanitary landfill near Dompoase, approximately 10km from Kwame Nkrumah University of Science and Technology (KNUST). Commissioned in January 2004, it was established as part of the Urban Environmental Sanitation Project (UESP I), which received funding from the World Bank and the Department for International Development. Africon and Conterra Company designed the landfill, which Zoomlion currently manages under the supervision of the Ministry of Sanitation and Water Resources (MSWR). The facility spans approximately 1000 acres. Until July 2019, it served as the destination for all industrial, institutional, commercial, and domestic waste generated within the Kumasi metropolitan area, the second-largest city in Ghana (Adjei-Boateng *et al.*, 2023). The landfill began operations in May 2004 with a design lifespan of 15 years, which has now been reached, and some sections have already been decommissioned.

3.2 Data Acquisition

An electrical resistivity tomography survey was conducted across the Geomembrane Liner cell at the landfill in three parallel profile lines (PL): PL 1, PL 2 and PL 3 with a 5m interval. (Fig 5) The geoelectrical data were collected using the AGI Supersting™. The device features 28 electrodes, enabling fully automated measurement of the subsurface apparent resistivity using the Schlumberger configuration on each profile line. The length of each profile line was 108m, with electrode spacing at a 4m interval, enabling investigation to a depth of approximately 17m.

Figure 5: Survey Profile Lines

To begin the survey, a measuring tape was laid, and the multi-electrode cable was laid out in a straight line across the profile line, with stainless steel electrodes inserted into the ground at regular intervals of 4 m. Each electrode was securely connected to the cable take-outs using

electrode clips, ensuring firm contact with the soil. The SuperSting™ device is positioned at electrode one, which marks the beginning of the array. It is connected to the electrode cable via a switch box. A LifePO4 battery is used to power the system, and the SuperSting™ is turned on for measurement.

Before initiating data acquisition, several critical settings were configured to ensure accurate and efficient measurements. First, the system was set to *Automatic Mode*, which enables the instrument to execute a preprogrammed survey sequence without manual intervention. Resistivity was selected.

Next, a command file was created and loaded. This file defines the electrode array type (Schlumberger), the electrode spacing (4m), the measurement sequence, and integration parameters. It also includes options for roll-along surveys, which allow for extended profiling by shifting the cable layout incrementally. The roll-along was not activated.

The survey file name was entered using the console interface on the main menu. This name was used to store all collected data and reflected the project for easy identification. The switch system was selected for Swift, and the cable address table was configured to match the number of electrodes per section.

Electrical parameters such as injection current and voltage limits were carefully set. These values control the strength of the current sent into the ground and protect the system from overvoltage conditions. Integration time and stacking cycles were also defined, balancing signal quality against acquisition speed.

Before starting the survey, the zero-check was run to determine whether the electrodes were firmly rooted in the ground. Finally, all settings were reviewed for consistency with the field layout and survey objectives. Once confirmed, the *Start* was pressed to begin automatic data collection. The data collected from the device was transferred to a computer for processing.

Figure 7: AGI SuperSting and LifePO4 battery

Figure 6: Inserting electrodes into the ground

3.3 Data Processing

Resistivity data were acquired using the SuperSting™ system configured with the Swift automatic switching system and a 28-electrode layout. This setup enabled efficient data acquisition across the survey lines without manual electrode selection. Data were processed using AGI EarthImager™ 2D inversion software to generate subsurface resistivity profiles.

Initial data filtering followed threshold parameters recommended by AGI (Advanced Geosciences Inc., 2020) and adapted to site-specific conditions. Measurements with voltage values below 0.2 mV and absolute voltage-to-current ratios (V/I) below 0.2Ω were excluded to minimise low-signal noise and instability. Apparent resistivity values were constrained between 0.5 and $10^5\Omega$ to eliminate outliers and non-physical readings. A repeatability threshold of 3% and a reciprocal error limit of 10% were applied to retain consistent and symmetric measurements. Additionally, every third data point was skipped to reduce redundancy and optimise inversion runtime. Negative resistivity values and isolated spikes were automatically removed, and the Keep All Data option was disabled to allow selective filtering.

Inversion was performed using the Smooth Model Inversion method, which applies a damped least-squares algorithm with spatial smoothing to produce a continuous resistivity distribution (Loke et al., 2003). The model geometry was defined with depth on the Y-axis and a vertical orientation set to positive upward. Electrodes were snapped to model nodes within 0.1m, and a horizontal distance scale factor of 2.5 was applied to account for terrain compression.

Forward modelling employed the Finite Element Method (FEM) for high spatial accuracy in complex subsurface geometries. The solver used Cholesky Decomposition, suitable for symmetric, positive-definite systems, and Dirichlet boundary conditions to fix potential values at model edges. The mesh was discretised with 5 divisions, and both the thickness incremental factor and depth factor were set to 1.1 for gradual refinement. A maximum of 100 conjugate gradient (CG) iterations was allowed, with a convergence criterion of 10^{-6} to ensure numerical stability.

Inversion control parameters were refined to balance smoothness, resolution, and data fidelity. The process was limited to eight iterations, with a root-mean-square (RMS) error threshold of 3% and a minimum error reduction of 5% between iterations. The L2 norm was selected as the objective function to minimise squared residuals. Stabilising and damping factors were both set to 10, consistent with AGI recommendations and prior field experience involving leachate zones and moisture gradients. Data weights were assigned based on estimated noise levels of 3%, and reciprocal error weighting was enabled. Noisy data were suppressed using robust data and model conditioners (both set to 1), and a resolution factor of 0.2 was applied to control the influence of poorly resolved regions.

The inversion began from a pseudosection-based starting model, with the final model type defined as Smooth Ohm-m, constrained within a resistivity range of 6 to 10^4ohm-m. Model parameter dimensions were set to 1m in both horizontal and vertical directions, with a horizontal-to-vertical roughness ratio of 0.5 to allow moderate lateral smoothing.

These combined processing and modelling steps produced a robust resistivity model suitable for delineating hotspot zones and stratigraphic transitions within the landfill body, following the methodology outlined by Frid *et al.* (2008) and Loke and Barker (1996).

3.4 Laboratory Investigation of Apparent Negative Resistivity Values

3.4.1 Aim

To investigate and confirm the underlying cause of negative resistivity values observed during ERT surveys, specifically attributing the anomaly to the influence of highly conductive materials situated near the receiving electrodes, which may act as secondary sources of current and distort the measured potential field.

3.4.2 Materials and Apparatus

- Soil Box: Used to contain the soil sample during resistivity measurements.
- Lateritic Soil: Serves as the test medium for electrical resistivity analysis.
- Salt (NaCl): Introduced to simulate ionic contamination and modify soil conductivity.
- Water: mixed with salt and soil to adjust moisture content and conductivity levels.
- Calibrated Volumetric Cylinder: Ensures accurate measurement of water.
- Resistivity Instrument (ABEM Terrameter): Measures the electrical resistivity of the soil sample.
- Connecting Wires: Link the resistivity instrument to the electrodes in the soil box.
- Stirring Rod: Used to homogenise the salt solution.
- Battery: Powers the resistivity instrument.

3.4.3 Procedure

To investigate the influence of highly conductive materials on electrical resistivity within landfill subsurface conditions, a controlled laboratory procedure was conducted.

- Measured 50g of salt and 1000ml (1 L) of water.
- Added the salt to the water in a calibrated cylinder and mixed thoroughly to form a saline solution.
- Grinded a portion of the lateritic soil sample and added a small amount of water to restore its natural moisture content.
- Poured the moistened soil into the soil box and divided it into two equal parts:
  - Portion A: Soil mixed with a small amount of water (natural moisture only).
  - Portion B: Soil mixed with the concentrated salt solution.
- Connected wires from the soil box electrodes to the resistivity instrument. One current electrode and one potential electrode were placed in the dry soil portion, while the other current electrode and potential electrode were positioned in the wet soil portion to simulate conductivity contrast across the test medium.
- Powered the instrument by connecting it to a battery.
- Took resistivity readings for both soil portions under controlled conditions.

Check the Appendix for pictures of the laboratory test.

4 Results and Discussion

4.1 Results

Inverted 2D resistivity sections for Profile Lines 1, 2, and 3, showing subsurface resistivity distribution across the decommissioned landfill site (Fig. 8, 9 and 10, respectively). Horizontal axes represent electrode positions spaced at 4 m intervals; vertical axes indicate depth in meters. The colour scale reflects resistivity values in ohm-m, with warmer colours denoting higher resistivity.

Table 2: Adopted resistivity values for interpretation

Material (Subsurface Section)	Resistivity (Ω·m)	References	Colour
Clay	3 – 150	Guerin et al. (2004); Hack (2000); Knieszel (2000)	
Saltwater-saturated clay	1.5 – 2	Abu-Zeid & Santarato (2004)	
Leachate	0.9 – 5	Guerin et al. (2004); Zume et al. (2006)	
Domestic garbage	12 – 30	Hack (2000)	
Scrap metal	1 – 12	Hack (2000)	
Sandy soil with clay	60 – 100	Reynolds (2011)	
Clayey sandy soil	30 – 60	Reynolds (2011)	
Mixture of leachate and waste	<10	Dahlin et al. (2010)	
Saturated clay (uncontaminated)	<10	Guérin et al. (2004)	
Unsaturated waste	>1000	Leroux et al. (2007); Dahlin et al. (2010)	
Unsaturated soil	>1000	Leroux et al. (2007)	
Saturated waste (without leachate)	30 – 150	Dahlin et al. (2010)	
Saturated soil (uncontaminated)	30 – 150	Guérin et al. (2004); Dahlin et al. (2010)	
Sandy or silty compacted soil	100 – 1000	Telford et al. (1990)	
Loamy compacted soil	30 – 300	Reynolds (2011)	
Organic-rich soil	50 – 500	Zailani et al. (2023); Dahlin et al. (2010)	
Clay-rich compacted soil	10 – 100	Guérin et al. (2004); Reynolds (2011)	

4.1.1 Laboratory Investigation of Apparent Negative Resistivity Values

Three resistivity values were recorded using the ABEM Terrameter system under controlled laboratory conditions, with electrodes placed across soil portions exhibiting varying moisture content. (Fig 8) The readings were as follows:

Table 3: Recorded Resistivity Values in the Laboratory

Moisture Condition in Wet Zone	Recorded Resistivity Value (kΩ)
Low moisture	104.47
Moderate moisture	100.97
High moisture	100.07

The results indicate a progressive decrease in resistivity as the moisture content in the wet zone increased.

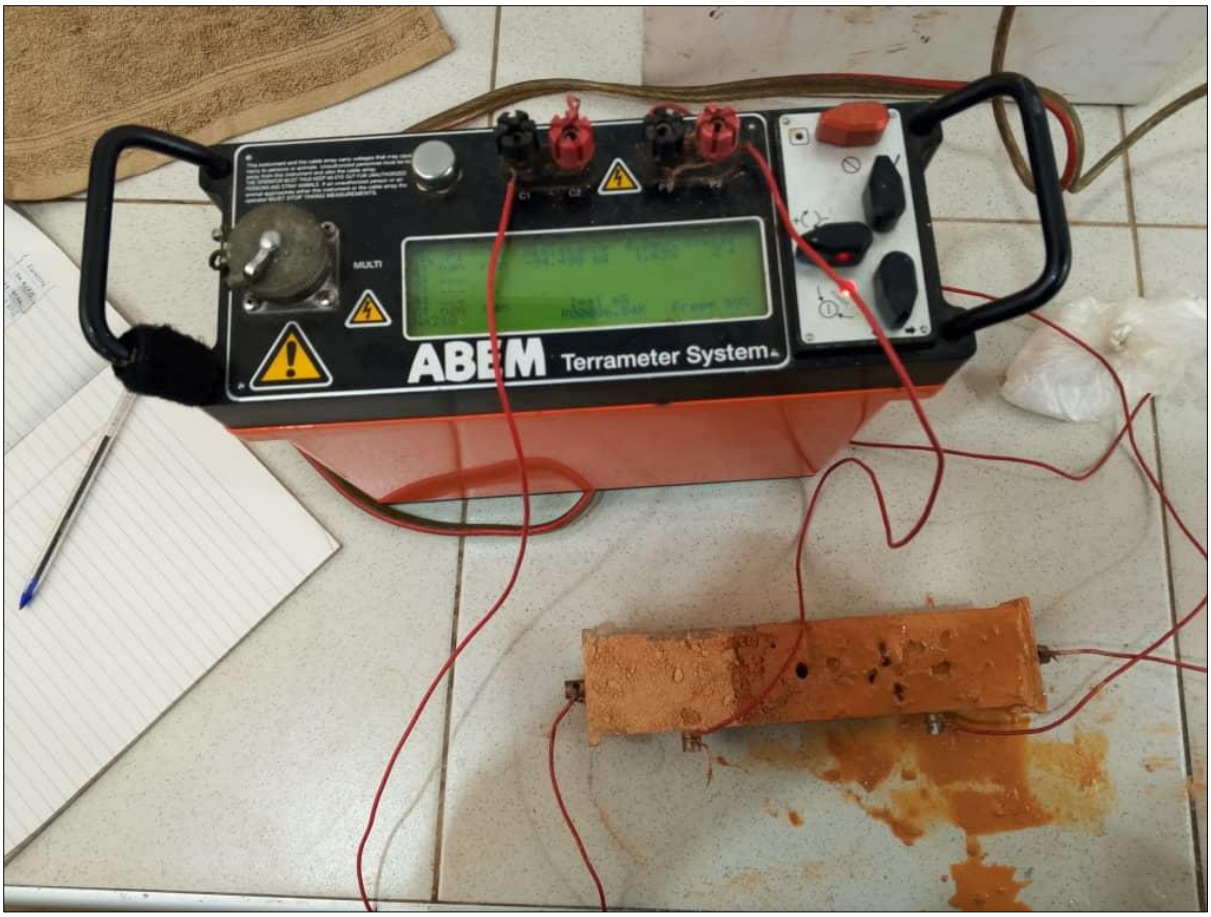

Figure 8: Laboratory setup showing Resistivity Instrument connected to a Soil Box via electrode wires

4.1.2 Treatment of Negative Resistivity and Flagged Values

Although apparent resistivity values are not conventionally used for direct interpretation in ERT, they were carefully considered in this study due to the unique geophysical characteristics of the site, a decommissioned landfill. As noted in prior surveys of similar environments (e.g., Soupios et al., 2005; Loke, 2010), negative resistivity values frequently arise due to high subsurface heterogeneity, buried metallic waste, or extreme conductivity contrasts. While these values are non-physical, they are often signal zones of anomalous behaviour that merit further scrutiny.

In this study, negative resistivity values were detected during data import into EarthImager and are visually represented as empty cells (square boxes), highlighted with red circles in Figure 9. During inversion, the software automatically excludes these values from computation. However, their spatial influence is retained through interpolation, as the inversion algorithm estimates values based on adjacent data points. Although the negative values are not directly inverted, their locations still influence the final resistivity distribution.

Given the landfill context, where subsurface anomalies are expected and may correspond to leachate plumes, buried waste, or moisture-saturated zones, the apparent resistivity data, particularly the distribution and clustering of negative values, provided valuable interpretive clues.

Similarly, flagged resistivity values were identified during data import and are visually represented as black cells, also circled in red in Figure 10. These flagged values were likewise excluded from inversion by the software, yet their spatial influence was preserved through surrounding interpolations. Although not directly inverted, their positions contributed to shaping the final resistivity model.

In both cases, the presence and clustering of anomalous values, whether negative or flagged, offered important insights into subsurface conditions. In the context of landfill geophysics, such anomalies may correspond to zones of extreme resistivity (either high or low), and their careful consideration enhances proper interpretation.

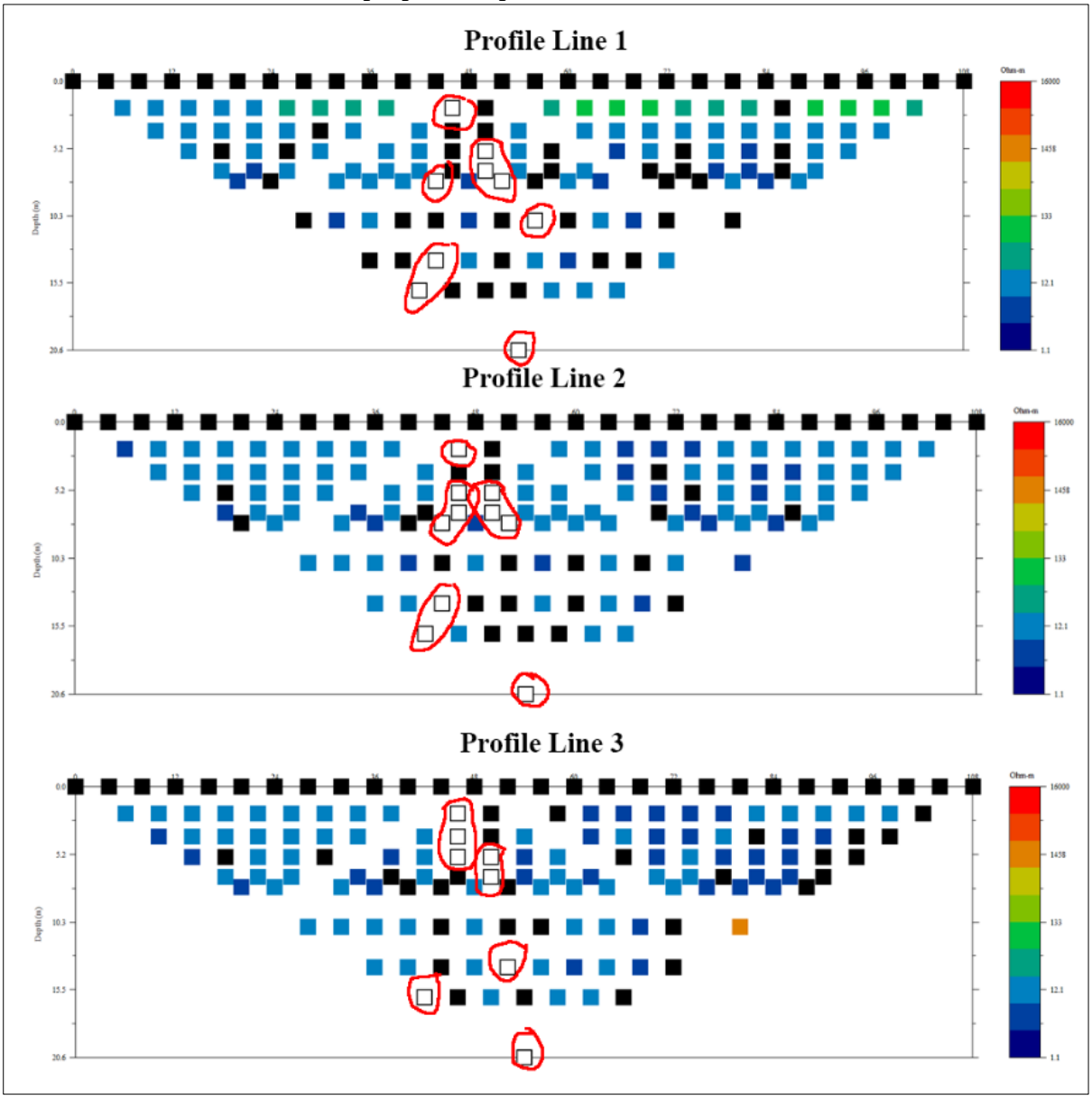

Figure 9: Apparent resistivity profile showing empty cells circled in red, representing negative values

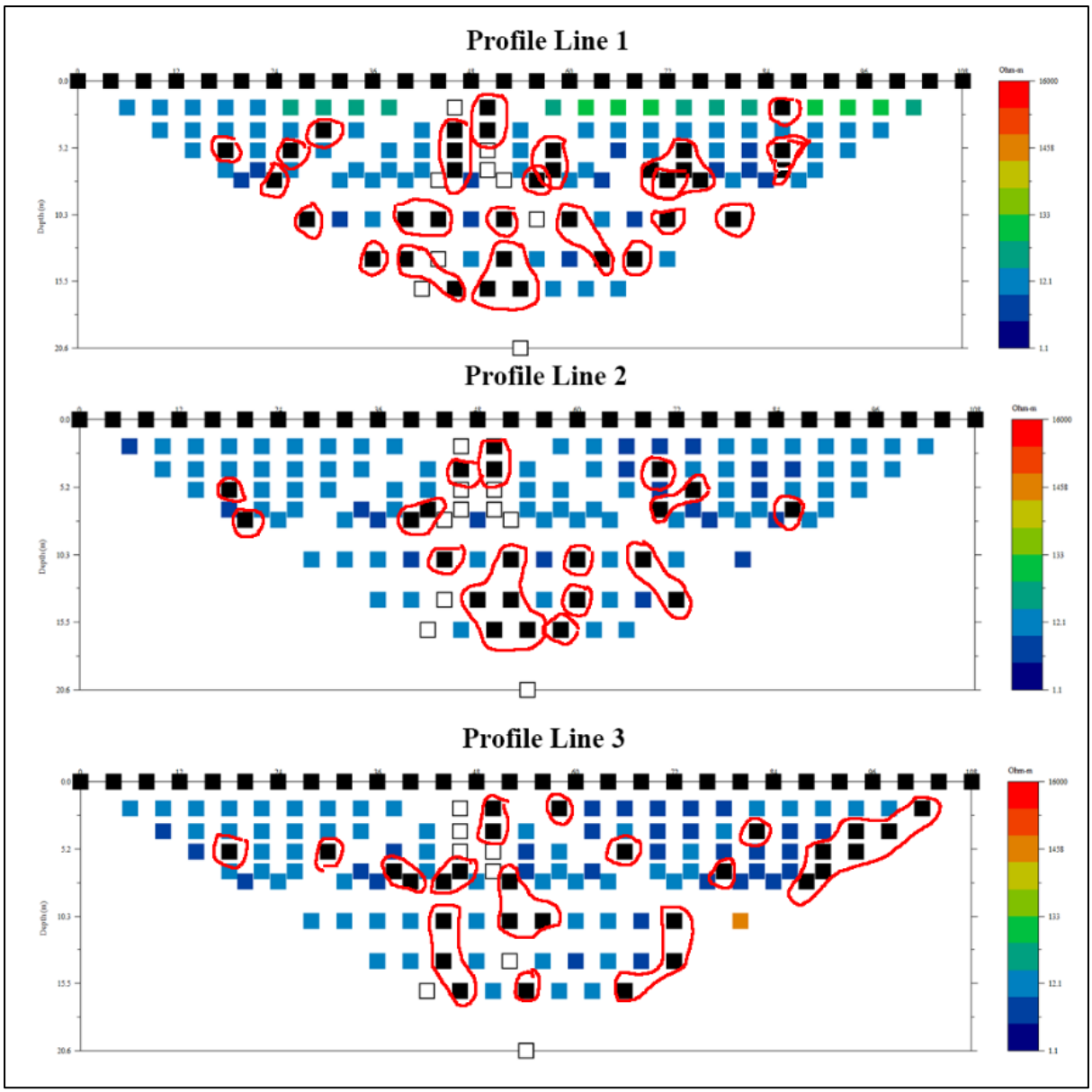

Figure 10: Apparent resistivity profile with black cells circled in red, indicating flagged values

4.1.3 Profile Lines 1, 2 and 3

Figure 11 presents the inverted resistivity of Profile Lines 1, 2 and 3. With profile line 1, the resistivity values from the measured data cover a wide range, from 2 to 16,000 Ω·m. However, 80% of the data shows resistivity values of less than 300 Ω·m. In the lower section, below a depth of 2m, the materials exhibit resistivity values ranging from 1.0 to 150 Ω·m. Although the resistivity below 2m is generally low, it consists of 40% of values between 1 and 10 Ω·m, 50% with values between 10 and 30 Ω·m and 10% with 30 and 150 Ω·m Above a depth of 2m, the resistivity values are recorded within the range of 1 to 16,000 Ω·m, with most falling between 100 and 2000 Ω·m, accounting for approximately 90% of the data. Additionally, there are a few pockets, about 2%, featuring very high resistivity values above the 2m depth.

For profile line 2, the measured data show a range of resistivity values from 2 to 16000 Ω·m. However, 94% of the data indicate resistivity values of less than 150 Ω·m, with only a few values ranging from 200 to 2000 Ω·m. The higher resistivity values are predominantly found at depths less than 2m, specifically between electrodes 22 and 26.

For profile line 3, the measured resistivity values range from 2 to 16000 Ω·m. However, 98% of the data reflect resistivity values ranging from 2 Ω·m to 50 Ω·m, with the predominant range

falling between 2 and 20 Ω·m. There is, however, a portion closer to electrode 25 that displays resistivity values higher than 50 Ω·m.

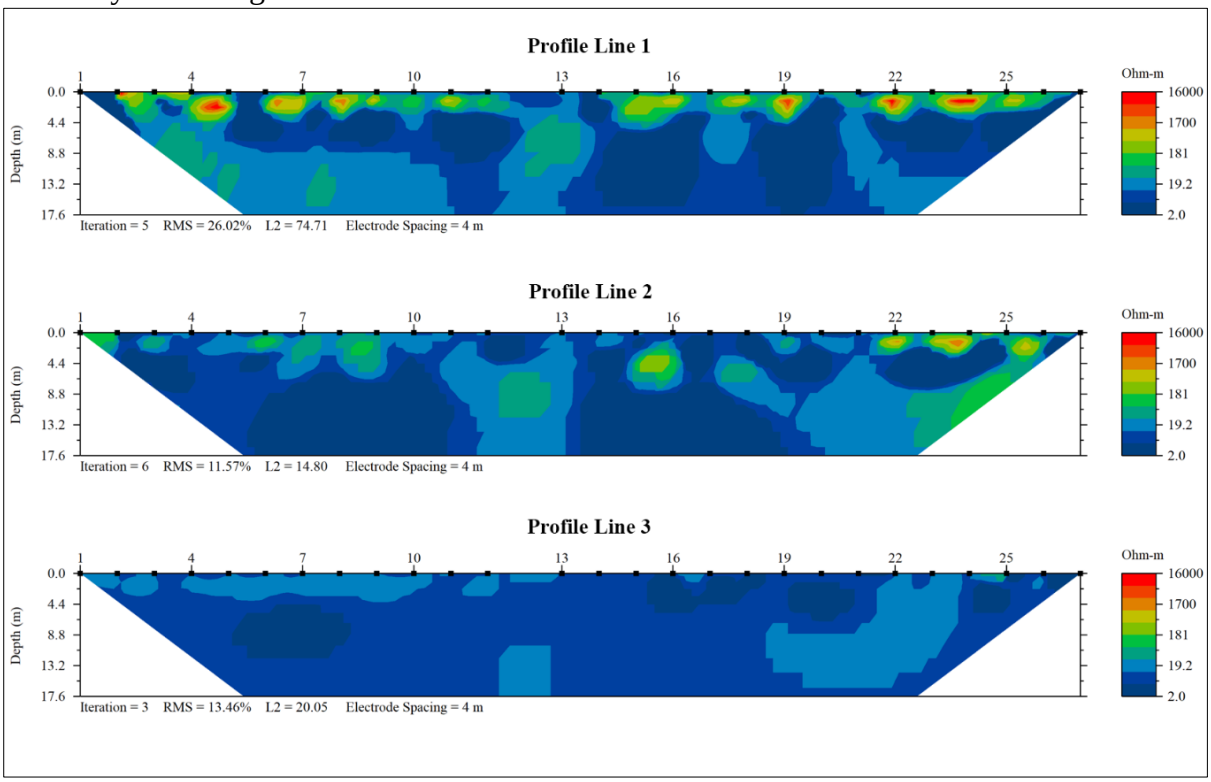

Figure 11: Inverted resistivity images of PL1, PL2 and PL3

4.2 Discussion

4.2.1 Laboratory Investigation of Apparent Negative Resistivity Values

The resistivity values recorded, 104.47 kΩ, 100.97 kΩ, and 100.07 kΩ, demonstrated a downward trend as the moisture content in the wet zone increased. This behaviour aligns with the expected response of saline-enhanced soils, where higher moisture and ionic concentration improve electrical conductivity and reduce resistivity.

Such conditions, particularly when highly conductive zones are situated near receiving electrodes, can distort the potential field measured during ERT surveys. This distortion may lead to anomalous outputs, including apparent negative resistivity values, which are non-physical but diagnostically significant.

4.2.2 Capping Layer

A gentle slope runs across the site from Profile Line 3 to Profile Line 1, influencing the deposition and thickness of the landfill covering. Field input from the site manager indicated that the landfill was capped with approximately 0.3m of soil. However, direct field observations revealed that sawdust had been visibly dumped across the surface, suggesting that the upper layer is not pure soil but rather a sawdust–soil mix, and in some areas, predominantly sawdust. (see appendix) The underlying compacted soil layer, whose thickness was not known before this study, was later interpreted through resistivity inversion to extend approximately 0.5m beneath the sawdust-soil mix. This discrepancy highlights the importance of integrating geophysical and visual evidence when characterising cover materials.

Due to gravitational settling and surface runoff, the sawdust–soil mix has redistributed downslope, resulting in a thicker cover at Profile Lines 1 and 2 compared to Profile Line 3.

The variation in the covering layer is clear in the inverted resistivity images across the three profiles. Profile Line 1, located downslope, shows a pronounced high-resistivity zone near the surface, indicating a thicker covering layer composed of compacted soil and a sawdust-soil mix

on top. Profile Line 2 exhibits a similar resistive signature, although it is slightly lower. This could be attributed to its intermediate slope position and partial accumulation of materials. In contrast, Profile Line 3, situated upslope, reveals a thinner resistive layer and significantly lower resistivity values, suggesting reduced surface cover and possible moisture accumulation. Typically, a thinner cover is linked to increased drying and desiccation, which can cause shrinkage and micro-cracking in the compacted soil. However, it may also allow upward moisture migration from the waste body below. The lack of sufficient overburden pressure or an effective capillary break can lead to leachate or gas-condensate rising into the capping layer, increasing saturation and reducing resistivity, a phenomenon consistent with unsaturated flow behaviour described by Hillel (1998). Therefore, the low resistivity observed at Profile Line 3 may indicate a localised area of moisture accumulation or proximity to waste, despite its upslope position. This situation highlights the complex interactions among slope geometry, material composition, and subsurface hydrological behaviour in landfill environments.

4.2.2.1 Thickness

From Figure 12, the annotated horizontal lines with filled circles on the left-hand side represent the estimated capping layer thicknesses across Profile Lines 1, 2, and 3. Based on resistivity contrasts in the inversion models and supported by field observations, these estimates indicate that the cover (compacted soil + sawdust-soil mix) ends at roughly 1.8m depth for profile lines 1 and 2, and approximately 1.2m for profile line 3. However, the interval between the surface and the first annotated line, the sawdust-soil mix, varies slightly among profiles, likely due to gravitational settling and surface runoff, which have unevenly redistributed the sawdust-soil mix along the slope. Based on resistivity patterns and material composition, the effective thickness of the compacted soil layer is interpreted to be approximately 0.5m, underlying the sawdust-soil mix.

As noted in Section *4.2.1*, Profile Line 3 exhibits a thinner resistive zone and lower resistivity values. This does not necessarily indicate an inadequate compacted soil layer but may reflect localised saturation caused by upward moisture migration from the waste body, facilitated by reduced overburden pressure. This interpretation builds on the capillary rise mechanism previously discussed, where insufficient vertical stress allows moisture to move upward through fine-grained soils with high pore connectivity (Hillel, 1998). Despite these spatial variations, all profiles display near-surface resistive zones that align with the expected electrical properties of engineered landfill caps.

The resistivity ranges observed across the profiles further support the interpretation of capping layer thickness. Profile Line 1, with values ranging from 100 to 16,000 Ω·m, reflects a dry and well-compacted cover, likely enhanced by downslope accumulation. Profile Line 2 shows moderately high values (200–2,000 Ω·m), consistent with intermediate cover thickness and moisture content. In contrast, Profile Line 3 exhibits significantly lower resistivity (10–50 Ω·m), suggesting localised saturation or upward moisture migration due to reduced overburden pressure.

4.2.2.2 Uniformity

Although all three profile lines exhibit similar vertical thicknesses for the compacted soil layer, the lateral uniformity of these layers varies across the site. In Figure 13, the resistive zones in Profiles 1, 2, and 3 show lateral inconsistencies, indicating that the compacted layer is not homogeneously distributed across the horizontal extent. The elliptical shapes annotated on the inverted images highlight zones where pore structures are inferred within the compacted soil layer. These features suggest variations in porosity and moisture retention, which may influence local resistivity values and contribute to the observed heterogeneity.

Across all three profile lines, zones between electrodes 13 and 16, and electrodes 19 and 22 consistently exhibit resistivity anomalies interpreted as pore structures within the compacted soil layer. These features, visible in the inverted resistivity images, suggest that despite the

overall uniform thickness of the compacted layer, there are microstructural discontinuities across the slope. The spatial alignment of these pore zones across Profile Lines 1, 2, and 3 indicates that they may be influenced by site-wide depositional patterns, compaction inconsistencies, waste settlement or subsurface moisture gradient.

Despite the observed lateral and cross-profile variability in resistivity, it is reasonable to infer that the same type of soil material was used for the compacted layer across all three profile lines. This assumption is supported by visual consistency in the soil matrix. However, the differences in electrical response stem from variations in compaction effort, moisture distribution, and microstructural heterogeneity.

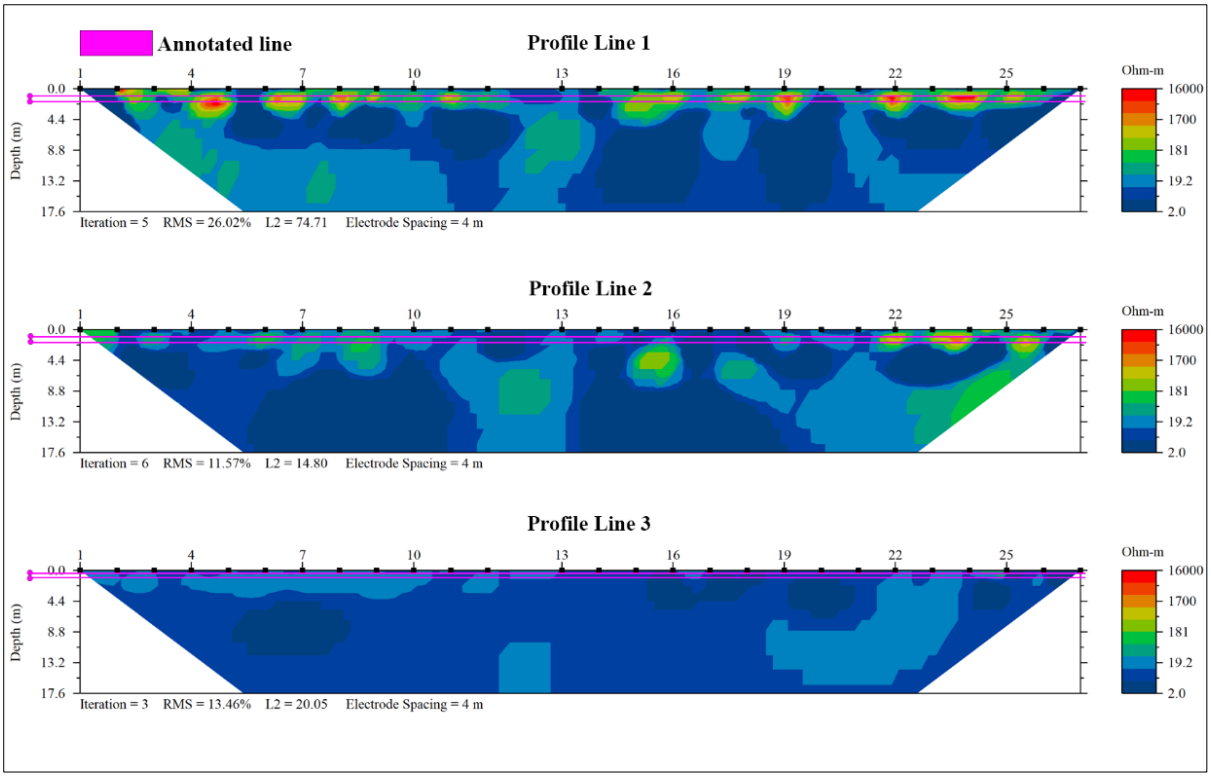

Figure 12: Inverted 2D resistivity sections for Profile Lines 1, 2, and 3, showing the thickness of the compacted soil

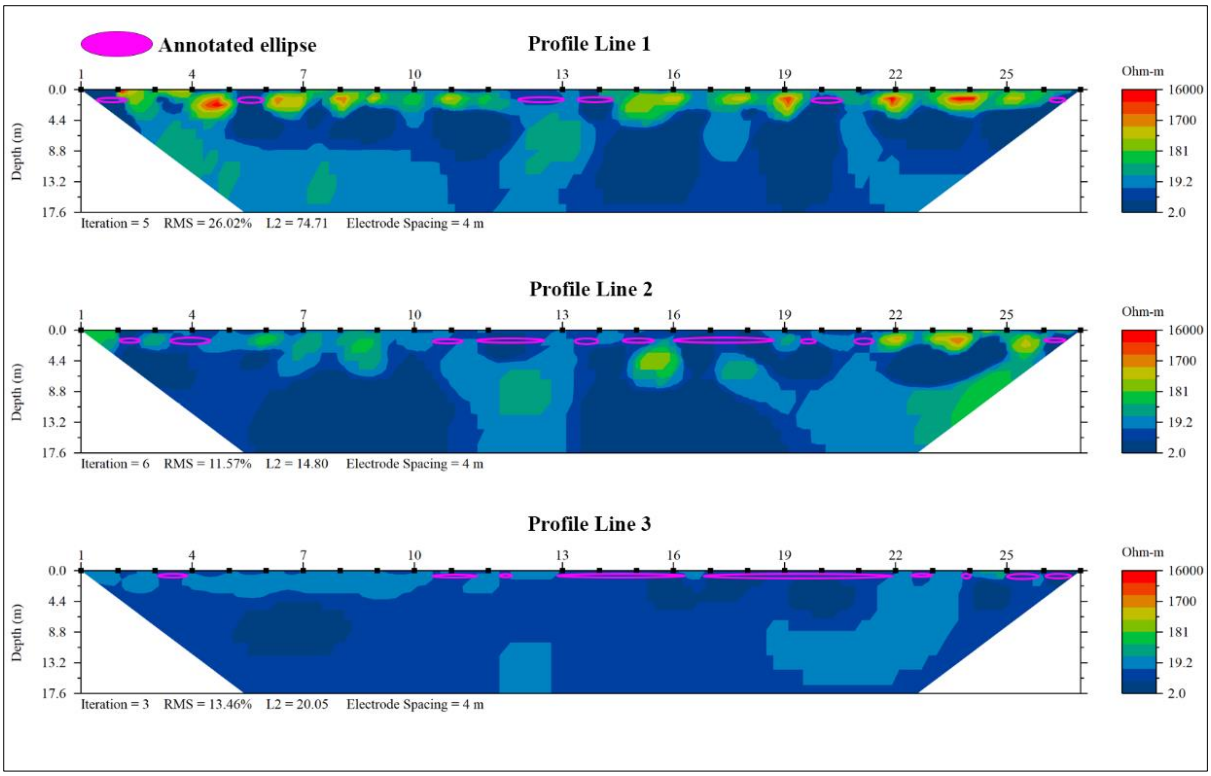

Figure 13: Inverted 2D resistivity sections for Profile Lines 1, 2, and 3, showing the pores in the compacted soil

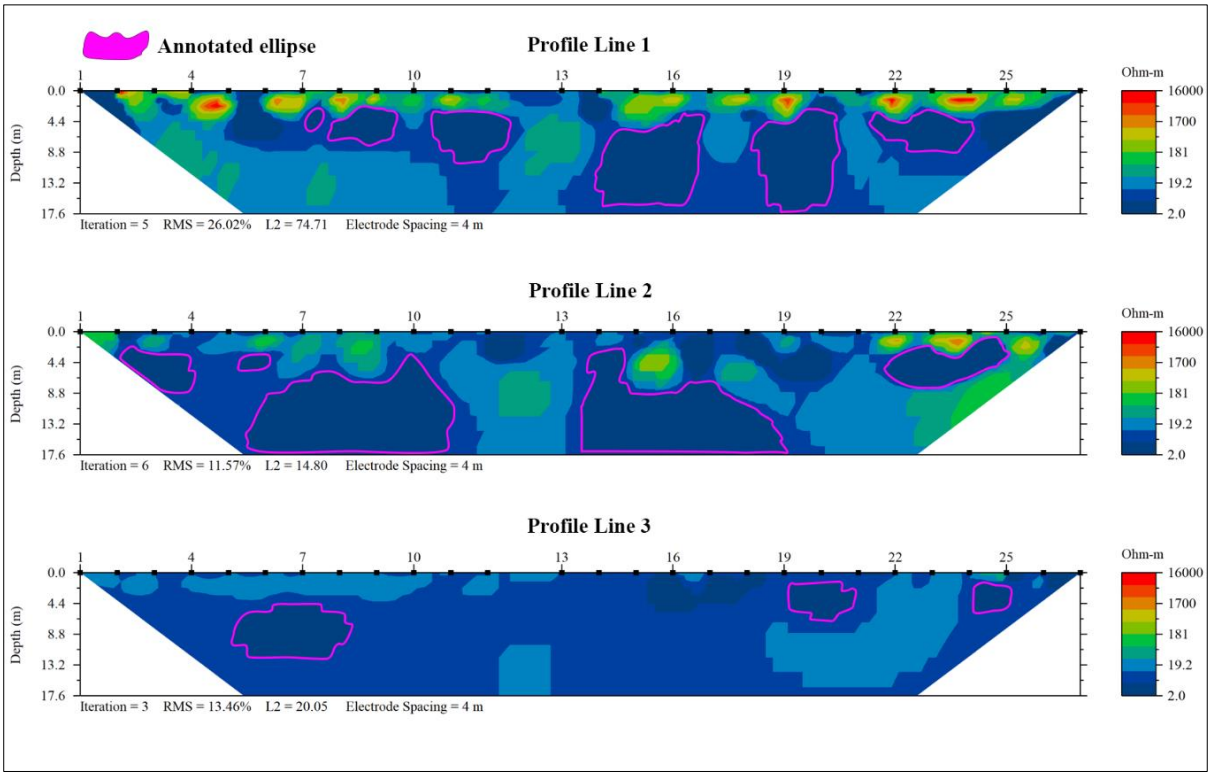

Figure 14: Inverted 2D resistivity sections for Profile Lines 1, 2, and 3, showing leachate zones

4.2.3 Leachate

The presence of pore zones and lateral heterogeneity within the compacted soil layer has broader implications beyond moisture retention. These microstructural discontinuities may compromise the hydraulic barrier function of the landfill cover, creating preferential pathways for fluid movement. Such pathways can facilitate downward, upward, or lateral migration of

water, which in turn promotes enhanced organic waste decomposition. This accelerated biodegradation process contributes to increased leachate generation. Although leachates typically exhibit low resistivity values (see Table 2), it is important to note that low resistivity zones in inverted images do not exclusively indicate leachate presence. Other factors, such as moisture accumulation, fine-grained saturated soils, or organic-rich zones, can also produce similar electrical responses, necessitating cautious interpretation. The annotated elliptical-like shapes in the inverted resistivity profiles in Figure 14 suggest zones of potential leachate accumulation or migration within the subsurface. These areas are characterised by low resistivity values, which are consistent with the electrical properties of leachates (see Table 2) due to their high ionic concentration and elevated moisture content. However, while the resistivity signatures align with expected leachate behaviour, it is important to note that low resistivity alone is not definitive. Similar responses may arise from saturated fine-grained soils, organic-rich zones, or moisture pockets unrelated to leachate movement. Therefore, these elliptical zones are interpreted as leachate-prone regions.

Even though certain subsurface zones exhibit similar resistivity values, their depth position significantly influences interpretation. Zones with low resistivity values closer to the surface are more likely to represent saturated soil, particularly within the compacted layer, where moisture accumulation is driven by surface infiltration or capillary rise. In contrast, deeper low-resistivity zones may reflect leachate migration, organic-rich decomposition zones, or naturally conductive materials, requiring careful contextual analysis.

5 Conclusion and Recommendation

5.1 Conclusion

This study demonstrated the effectiveness of ERT in evaluating the integrity of the capping layer and identifying leachate-prone zones within the decommissioned Oti Landfill. Through 2D resistivity profiling and inversion modelling, the thickness and lateral uniformity of the capping layer were successfully delineated, with high resistivity zones near the surface indicating well-compacted cover material. The resistivity models also revealed subsurface anomalies consistent with leachate accumulation, supporting the identification of environmental hotspots.

To enhance interpretive accuracy, a resistivity classification table was adopted to guide subsurface characterisation and anomaly detection. This non-invasive approach provided a cost-effective and spatially resolved framework for landfill assessment, reducing the need for intrusive sampling and improving diagnostic confidence.

While this study did not directly assess groundwater or surface water contamination, the delineation of leachate-prone zones has implications for environmental protection, particularly in urban settings where residential proximity heightens the risk of subsurface migration. These findings offer actionable insight for post-closure monitoring, targeted remediation, and future site redevelopment.

In addition, the laboratory investigation demonstrated that increasing moisture content in saline-enhanced soils leads to a consistent decrease in resistivity, confirming the expected conductivity behaviour. More importantly, the setup successfully replicated conditions under which apparent negative resistivity values may arise. This validates the hypothesis that electrode proximity to highly conductive zones can distort the potential field, resulting in inversion artefacts and anomalous outputs during ERT surveys. These findings underscore the need for careful electrode placement and interpretation protocols in resistivity-based diagnostics, especially in heterogeneous or moisture-variable environments.

Although one of the study's objectives was to determine the depth of the landfill liner, the inversion models did not penetrate sufficiently to resolve this boundary. This limitation reflects the depth constraints of the current array configuration and informs the scope of future investigations.

5.2 Recommendation

To build on the findings of this study, future research should explore the integration of geophysical data with hydrologic and geochemical models to simulate contaminant transport and evaluate cap performance under variable climatic conditions. Such interdisciplinary approaches can improve predictive capacity and inform adaptive management strategies for landfill sites.

To overcome the depth limitation encountered in this study and improve subsurface characterisation, future surveys should consider deploying extended electrode arrays alongside complementary geophysical methods such as Electromagnetic Induction (EMI) and Induced Polarisation (IP). These techniques, when integrated with Electrical Resistivity Tomography (ERT), can enhance vertical and lateral resolution, improve anomaly discrimination, and support multi-parameter decision frameworks for sustainable landfill management.

Post-closure monitoring protocols should incorporate periodic geophysical surveys, such as ERT, to track changes in leachate migration and capping layer integrity. These assessments can guide targeted remediation efforts, reinforce public health safeguards, and support the safe reuse of decommissioned landfill sites, especially in urban contexts where exposure risks are elevated.

Finally, further laboratory simulations are recommended to refine the understanding of inversion artefacts under varying moisture and salinity conditions. This will improve field

interpretation accuracy and support the development of robust diagnostic protocols for heterogeneous landfill environments.

6 Reference

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7 Appendix

Figure 15: Vehicle dumping sawdust on the site